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MAGNETOSTRATIGRAPHY OF THE TOARCIAN AMMONITICO ROSSO LIMESTONE AT IZNALLOZ, SPAIN

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Abstract. The Iznalloz section (Betic cordillera, Spain) provides a continuous outcrop (20 m thick) of Toarcian ammonitico rosso limestones with all the ammonite zones defined in the Betic realm. The characteristic magnetization direction for each sample was obtained by detailed thermal demagnetization and was stable from about 450 to 560°C. The magnetic polarity sequence extends from the Polymorphum to the Aalensis ammonite zone, and covers the whole Toarcian stage. Thus this section presents a good pattern of the reversal succession during Toarcian times. Considering hiatuses of sampling and excluding single-sample magnetozones, there does not seem to be any important change in the reversal rate during this period.

Introduction

Magnetostratigraphy has become an essential correlation and dating tool, especially for upper Cretaceous and Cenozoic sediments. This tool is not yet available for Jurassic and older sediments, because the magnetic polarity sequence of these periods is not well known. The Jurassic and Triassic polarity succession must be established by magnetostratigraphic studies of sedimentary sections, since there is no oceanic crust older than middle Jurassic.

On the other hand studies of magnetic reversal rates through geologic time allowed some geodynamic models to be proposed. For exemple the controversial model for coupling of core and mantle convection suggests that times of increasing reversal rates correspond with periods of fast true polar wandering (Courtillot and Besse, 1987). This model requires an increase in reversal frequency during the upper Triassic and early Jurassic times.

The magnetostratigraphic data available for the lower Jurassic are scarce (Márton èt al., 1980; Horner and Heller, 1983; Channell et al., 1984; Galbrun et al., 1988 a, b) and do not provide a complete magnetic polarity sequence for this period, nor a precise correlation with the ammonite biozonations developed in different biogeographic realms. This objective requires the study of many sedimentary sections given the difficulties of magnetostratigraphic studies (inconsistent preservation of primary remanent magnetization in relatively old sedimentary rocks, difficulty of collecting samples in marly facies, hiatuses in outcrops, frequent lack of fauna and difficulties in correlating ammonite biozonations from different realms).

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The Toarcian, the last stage of the Lias, extends from about 180 to 187 Ma on the time scale of Odin (in press) and could correspond to a high reversal frequency period (Courtillot and Besse, 1987). Despite several magnetostratigraphic studies of Toarcian age sediments (Horner and Heller, 1983; Channell et al., 1984; Galbrun et al., 1988 a,b) the magnetic polarity sequence for this stage is not well established. This is due mostly to difficulties of correlation between the different sections studied.

We present in this paper the first magnetostratigraphic results obtained from the Toarcian of Betic Cordillera (Spain) in the Tethyan realm. The results obtained provide an excellent record of the geomagnetic reversal history during the Toarcian.

Geological setting and stratigraphy

The studied section is located in the Betic Cordillera 3 km west of the village of Iznalloz along the Granada-Moreda railway (37.39° N, 3.49° W).

Tertiary tectonics created folding, lateral shear and terranes; thus the bedding in the Iznalloz section is overturned. Detailed ammonite biostratigraphy in the Toarcian of the Betic Cordillera has been established previously (Jimenez and Rivas, 1979). All the ammonite zones defined for the Toarcian are to be found in the Iznalloz section. A detailed description of this section has been published by Braga et al. (1981) and we will only present a brief summary of the lithostratigraphy of the Toarcian which comprises 20 m of ammonitico rosso limestones (see lithologic log in Figure 4).

The lower part of the Polymorphum ammonite zone is a prominent grey micritic limestone; the upper part of this Polymorphum zone and the Serpentinus and Bifrons zones are composed of red or grey marly limestones and red marls. The Gradata, Fallaciosum, Reynesi and Aalensis zones are mainly red nodular limestone interbedded with some marl layers. The nodular limestones represent a lower sedimentation rate than the marls.

Thus, a complete Toarcian section of ammonitico rosso limestones, which are good recorders of magnetic polarity reversals (Channell et al., 1982; Lowrie and Heller, 1982; Channell et al., 1984; Freeman, 1986) is present at this locality.

Magnetic properties

The section was intensively sampled with a mean sampling spacing of 13 cm, except in the marls of the lower part of the section where only a few samples could be cored. A total of 101 stratigraphic levels were sampled from the 20 m of this section. All measurements were made on a three-axis RS-01 (LETI/CEA) cryogenic magnetometer; the noise level is such that magnetization intensities of less than 2×10^{-5} A/m cannot be precisely measured. The natural remanent magnetization (NRM) of the Iznalloz limestones ranges from 3.3×10^{-5} to 2.6×10^{-3} A/m. The

lower part of the section exhibits weaker magnetization, the NRM intensity being about 0.2-5x10^{-4} A/m, while the upper part shows an increase of NRM intensity towards the top of the section up to 2.6x10^{-3} A/m in the uppermost bed.

Most NRM directions are of normal polarity (before any tilt correction) and group towards the NW. The ammonitico rosso limestones usually have a magnetic mineralogy dominated by magnetite (which is most probably detrital in origin), and authigenic haematite, grown very soon after deposition (Channell et al., 1982; Lowrie and Heller, 1982; Freeman, 1986). The acquisition of isothermal remanent magnetization (IRM) is a convenient way to characterize the magnetic carriers of pelagic limestones (Lowrie and Heller, 1982; Galbrun and Butler, 1986). IRM acquisition curves from the ammonitico rosso limestones of Iznalloz (Figure 1) indicate a rapid increase in applied field up to 0.1 or 0.5 T and a subsequent slow increase, saturation being unattained at 1.6 T. Progressive thermal demagnetization of the maximum IRM indicates a blocking temperature of 580°C (Figure 1, sample IZ 7A) and a small persistent component above 600°C which is clearly evident in sample IZ 9A. These characteristics show that the predominant low-coercivity mineral is magnetite and the high coercivities are due to haematite with high blocking temperature.

Twenty pilot samples representative of each lithology underwent progressive thermal demagnetization in 50°C increments from 100 to 500°C and in 20°C increments above 500°C. The directions of the principal components of magnetization in each sample were determined by least squares analysis using the software of Torsvik (1986). Two typical thermal demagnetization plots are shown in Figure 2, they both indicate two distinguishable components. A first, soft, component is removed by heating to 150-200°C. It has the present earth's field direction (before tilt correction) and thus is probably a viscous remanent magnetization (VRM). Above these temperatures a stable component is progressively removed. This has a blocking temperature spectrum ranging from 300°C to more than 560°C and is likely due to the magnetite as shown up by the IRM measurements. This component can be either of reversed polarity (sample 34-1) or of normal polarity (sample 99-1) and thus is most probably of primary origin. Above 560°C the remanent magnetization is too weak to be measured and mineralogical transformations occur probably because the samples exhibit considerable magnetic viscosity.

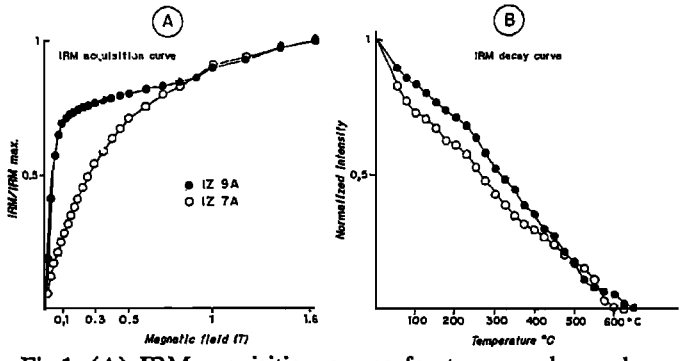


Fig.1. (A) IRM acquisition curves for two samples and (B) continuous thermal demagnetization of the maximum IRM acquired in a field of 1.6 T (see explanation in text).

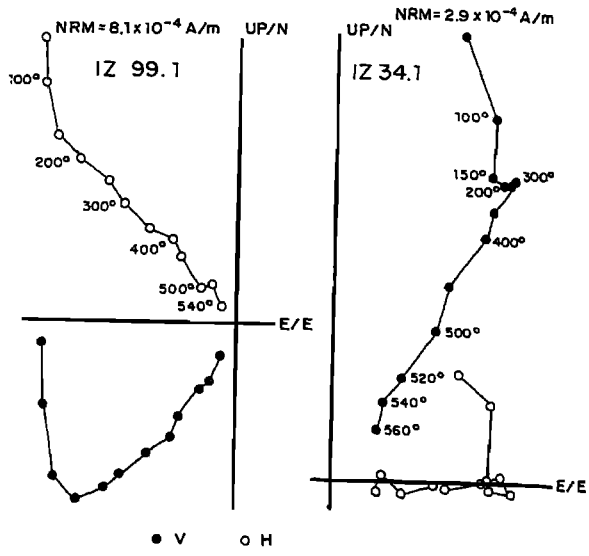


Fig.2. Orthogonal projections of thermal demagnetization data for two typical samples. The demagnetization range is given in centigrades, the horizontal (H) and vertical (V) projections are indicated. All directions are corrected for tilt (see explanations in text).

Given these results, the remaining samples from the Iznalloz section were measured at a selected set of about 4 or 5 demagnetization steps between 450 and 560°C. The characteristic remanent magnetization (ChRM) direction for each sample was averaged by a three-dimensional least squares technique (Kirschvink, 1980). These ChRM vectors form a nearly antipodal group of normal and reverse polarity directions, with some samples exhibiting intermediate polarity (Figure 3). The large scatter between directions is probably due to inadequate bedding corrections in places. As a matter of fact the section is sometimes disturbed and the dip of beds is difficult to estimate, especially in marly facies. Mean directions were calculated for both the normal and reverse polarity groups, and for all samples combined after exclusion of intermediate polarity samples (Table 1). The normal and reversed means are

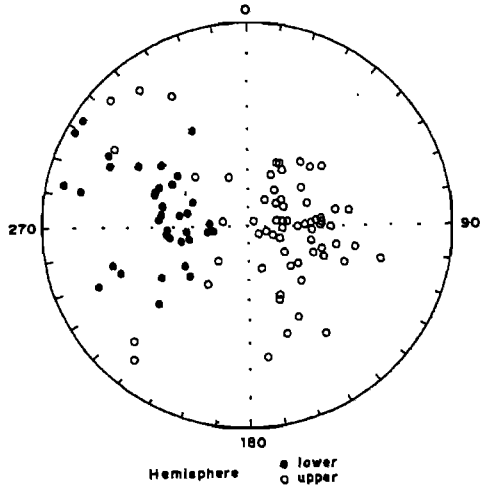


Fig.3. Stereographic plot of high blocking temperature component magnetization directions (after tectonic correction).

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TABLE 1. Characteristic mean magnetization directions (after tilt correction).

Samples	N	D	I	α_{95}	k
normal	33	272.9	43.5	7	12.1
reversed	47	84.9	-62.1	5.1	16
combined	80	269	54.6	4.5	12.1

statistically different; this is probably due to an incomplete separation of components, and may reflect a partial overprint by a late secondary component with a high blocking temperature.

The Iznalloz section is of overturned bedding and thus the mean paleomagnetic direction has no geodynamic significance.

Magnetic stratigraphy - Conclusion

The declination and inclination of characteristic magnetization directions plotted as functions of stratigraphic position, give a clear magnetic polarity sequence (Figure 4). Most of the polarity intervals are documented by several samples, those defined by single samples always present a good paleomagnetic direction without any doubt about the polarity, except one sample at the top of the Fallaciosum zone. The magnetic polarity sequence established at Iznalloz, which covers all of the Toarcian apart from a few interbeds in the middle part of the section in marly facies, is probably one of the most representative polarity successions for the Toarcian time. We may take note of the following with regard to this:

(1) Magnetostratigraphic studies of the Toarcian type sections of Thouars and Airvault (France) yielded two reliable magnetic polarity sequences (Galbrun et al., 1988 b), but some lacunae occur at the base of these condensed type sections (6.60 m thick for the Toarcian) and the magnetic polarity sequences established are probably not complete.

(2) The Toarcian of the Sierra Palomera section (Iberian Cordillera, Spain) is thick (75 m of limestones and marly limestones), but some stratigraphic or sampling gaps and difficulties in separating primary magnetization from secondary magnetizations hinder the establishment of a complete record of the magnetic polarity zones (Galbrun et al., 1988 a).

(3) The Valdorbia section (Central Italy) has poor faunal content and a major hiatus is suggested in this section (Guex, 1975; Géczy, 1984). Thus the magnetic polarity succession proposed for this section (Channell et al., 1984) is probably not complete.

It is difficult to correlate the Iznalloz polarity sequence to other reliable sequences such as the one established for the Breggia Gorge section (southern Switzerland) (Horner and Heller, 1983). This requires establishment of good correlations between ammonite biozonations in the different biogeographic realms and study of histories of sedimentation rate in different sections. Such a work needs long discussion and could not be presented in the present paper. Meanwhile it seems that there are more reversals in the Iznalloz section, especially in the upper Toarcian. This section,

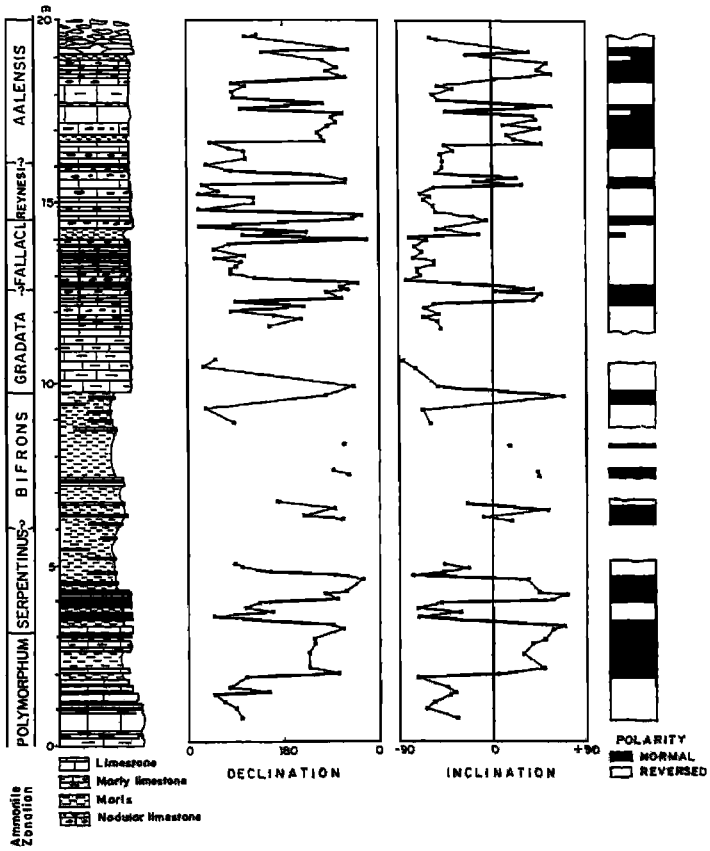


Fig.4. Ammonite zonation, lithological log and magnetostratigraphy (inclination and declination of the stable characteristic magnetization, magnetic polarity) for the Iznalloz section.

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which certainly has a good stratigraphic completeness, presents probably the best record of the geomagnetic reversal succession during the Toarcian. Excluding single sample magnetozones in the uppermost Toarcian (Aalensis zone), it does not seem that there is important variation in reversal rate during the Toarcian. This rate averages 4 reversals per ammonite zone. Meanwhile the sampling interval is less close in the lower part of the section (in part because of the marly facies) and other shorter polarity zones may exist.

During Jurassic times high reversal rates (5 to 10 reversals per ammonite zone) are known for the Bajocian and Bathonian (Steiner et al., 1987), and Pliensbachian (Horner and Heller, 1983). The magnetic polarity time scale for the lowermost Jurassic (Sinemurian and Hettangian) is not well documented but the few data available indicate a low reversal rate (Haq et al., 1987). The Aalenian of the Breggia Gorge section also shows a low reversal rate (Horner and Heller, 1983). The Callovian and Oxfordian polarity patterns are poorly documented, but the available results also indicate a moderate rate of magnetic reversals (Ogg and Steiner, 1988). We showed in this paper and previous work that the Toarcian has a moderate reversal rate. Thus the Jurassic exhibits periods of high reversal rate (Pliensbachian, Bajocian-Bathonian) alternating with times of low reversal rate (Hettangian-Sinemurian, Toarcian-Aalenian, Callovian-Oxfordian).

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